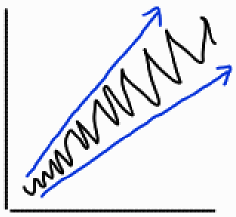
stationarity

① constant variation?

- no → variance stabilising transformation
- yes → no transformation



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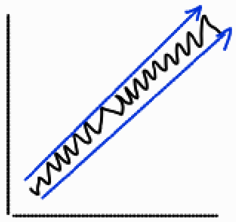


$$w_t = \begin{cases} \log(y_t) & \text{if } \lambda = 0 \\ (y_t^{\lambda} - 1)/\lambda & \text{otherwise} \end{cases}$$

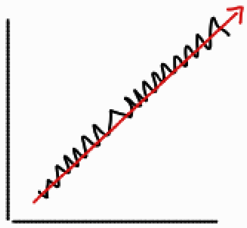
λ? Box-Cox transformation in R:

```
lambda → BoxCox.lambda(ts)
lambda
ts_lam → BoxCox(ts, lambda)
```

- ⊛ If $\lambda = 1$, no variance stabilizing needed
- ⊛ If $\lambda \approx 0.25$, $w_t \approx \sqrt[4]{y_t}$
- ⊛ If $\lambda \approx 0.33$, $w_t \approx \sqrt[3]{y_t}$
- ⊛ If $\lambda \approx 0.5$, $w_t \approx \sqrt{y_t}$

② constant mean?

- yes → no transformation
- no → detrend / deseasonalize via differencing



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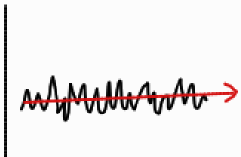

a) detrend?

e.g. $w_2 = y_2 - y_1$, $w_3 = y_3 - y_2$ ...

$w_t = y_t - y_{t-1}$, $t = 2, 3, \ldots, T$

elements

[ndiffs(ts_lam)] = 1 : requires first-order differencing at lag 1

[ts_lam_dif ← diff(ts_lam, 1)]

b) deseasonalize?

e.g. $w_{13} = y_{13} - y_1$, $w_{14} = y_{14} - y_2$ ...

$w_t = y_t - y_{t-g}$, $t = g+1, g+2, \ldots T$

elements

[nsdiffs(ts_lam)] = 1

requires first-order differencing at lag g (e.g. g = 12 for yearly seasonality of monthly observations):

[ts_lam_dif ← diff(ts_lam, 12)]

↓

final time series: ts_lam_dif

③ Is it stationary now?

a) KPSS test on stationarity of the mean:

[ur.kpss(ts_lam_dif)]

b) Check the residuals:

[checkresiduals(ts_lam_dif)]